CURRICULUM VITAE

PERSONAL INFORMATION

Name	Dr. Sahaya Vijay Jeyaraj
Email address	sahayavijay.j@gmail.com
Mobile No.	+91-7904355305, +81-8072023767
Date of Birth	06th February, 1993
Nationality	Indian
Languages Known	English, Tamil



EDUCATIONAL DETAILS

1. **Sep 2020 - Apr 2024**: **Ph.D.** from Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology (VIT University), Vellore, Tamil Nadu, India (Supervisor: Dr. Roy Santiago)

   Thesis Title: *A Study on Molecular Topology of Certain Bipartite Chemical Structures.*

2. **July 2016 - Sep 2017**: **M.Phil.** (Mathematics) from St. Joseph's College, Tamil Nadu, India (Supervisor: Dr. Koil Raj)

   Thesis Title: *Minimum Metric Dimension of Certain Networks.*

3. **June 2014 - Apr 2016**: **M.Sc.** (Mathematics) from St. Joseph's College, Tamil Nadu, India (Supervisor: Dr. Karuppiah)

   Thesis Title: *Vector space of graph.*

4. **June 2011 - Apr 2014**: **B.Sc.** (Mathematics) from Arul Anandar College, Madurai, Tamil Nadu, India.

RESEARCH EXPERIENCE

1. **April 2026 - Present: NIMS Postdoctoral Researcher** from Mathematical Catalysis Research Team, GREEN Center at National Institute for materials Science, Namiki Campus, Tsukuba city, Ibaraki, Japan. (PI: Dr. Hideshi Ooka)
2. **Jan 2025 - March 2026: Postdoctoral Researcher** from CSRS Biofunctional Catalyst Research Team, RIKEN National Research Institute, Wako campus, Saitama, Japan. (PI: Prof. Ryuhei Nakamura)

TEACHING EXPERIENCE

1. **Apr 2021- Feb 2024**: **Assistant Professor Fellowship** at the VIT University, Vellore, Tamil Nadu, India.
2. **Aug 2019- Sep 2020**: **Assistant Professor** at the Department of Science and Humanities, Vaigai College of Engineering, Anna University, Tamil Nadu, India.
3. **Jul 2017- Dec 2018**: **Assistant Professor** at the St. Joseph University, Nagaland, India.

PROJECT

1. Working in the "**FOREST project**" with Mathematical Catalysis Research Team, National Institute for Materials Science, Namiki Campus, Ibaraki, Japan.
2. Working in the "**Chemistry of the CO world project**" with CSRS Biofunctional Catalyst Research Team, RIKEN, Wako Campus, Saitama, Japan.

AWARD/ACHIEVEMENT

1. **Raman Research Award** for April 2023
2. **Raman Research Award** for June 2023
3. **Raman Research Award** for August 2023

RESEARCH ACHIEVEMENT

1. Obtained Amplification Rate of Autocatalysis in the Field of Origin of Life.
2. Obtained A New Mathematical Equation to Fit EIS in the Field of Electrochemistry.
3. **RARS Application Number:** R25007251

   Publication Type: Presentation

   Title: Calculating the lifetime of Autocatalytic Chemical Reaction Networks

   Authors: **Jeyaraj Sahaya Vijay,** Nakamura Ryuhei, Ooka Hideshi, Kawasaki Mayu

   Year: March 2026
4. **RARS Application Number:** R25007252

   Publication Type: Presentation

   Title: Timescale Decomposition of Linear Chemical Reaction network

   Authors: Ooka Hideshi, **Jeyaraj Sahaya Vijay**, Nakamura Ryuhei, Kawasaki Mayu

   Year: March 2026
5. **RARS Application Number:** R25007253

   Publication Type: Presentation

   Title: Stability Analysis of chemical reaction networks based on the amplification rate of autocatalysis

   Authors: Ooka Hideshi, **Jeyaraj Sahaya Vijay**, Nakamura Ryuhei, Kawasaki Mayu

   Year: March 2026

SKILLS

1. **Languages:** Strong reading, writing, and speaking competencies in English and Tamil.
2. **Programming Languages:** Python, C++, Java, HTML, CSS
3. **Software:** MATLAB, Chemcraft, Mathematica, LaTeX, Flash8.
4. **Online Certificate Course**: Python Programming for Machine learning (Certificate No: 1BHGBFVZ)
5. **Software Developed**: Python Software Packages - PyTIE_D, PyTIE_DS, PyTIE_SMS_DE, PyTIE_SMS_DSE (Git Hub link: https://github.com/sahayavijay/PyTIE-0.0.1.git)

PUBLICATIONS

1. **Sahaya Vijay Jeyaraj**, Hirokazu Komatsu, Tomoharu Suda, Ryuhei Nakamura, Hideshi Ooka, Kinetic Boundary between Autocatalysis and Diffusion in the Origin of Life, *ChemRxiv*, 2026. **DOI**: 10.26434/chemrxiv.15004763/v.
2. F. Sherin Rebecca, R. Sharmila, **Sahaya Vijay Jeyaraj**, Fikadu Tesgera Tolasa, Rajamanickam Pon Nivedha, Janani Ezhilan, Tony Augustine, Anticancer Activity with Physical Property Analysis of Mollugo Cerviana (L). Ser Whole Plant Extract on Hepatocellular Carcinomic Cell Lines -HEPG2, *Scientific Reports*, 2026. **DOI**: 10.1038/s41598-026-58687-5.

3. S. Prabhu, **Sahaya Vijay Jeyaraj**, V. Manimozhi, Micheal Arockiaraj, Roy Santiago, J. Celin Fiona, Krishna Balasubramanian, Topological, Graph-Spectral Properties, NMR and ESR Spectral Patterns of Hydrogen-Bonded Nanosheets of Boric Acid, *Journal of Cluster Science*, 37(48), 2026. **DOI**: 10.1007/s10876-026-02996-2.
4. K. Jyothish, Roy Santiago, P. Chandra Sekar, K. B. Gayathri, **Sahaya Vijay Jeyaraj**, Tony Augustine, Machine Learning Approach on Topological Features of Bottom-up Synthesized Graphene Nanoribbons using Valency Based Descriptors, *Frontiers in Chemistry*, 14, 2026. **DOI**: 10.3389/fchem.2026.1750413.
5. **Sahaya Vijay Jeyaraj**, Roy Santiago, Govardhan S., Tony Augustine, Jyothish K., PyTIE: A Python Program for the Evaluation of Degree-Based Topological Descriptors and Molecular Entropy, *arXiv*, 2025. **DOI**: 10.48550/arXiv.2510.26813.
6. Gayathri K. B., Roy Santiago, **Sahaya Vijay Jeyaraj**, Chandra Sekar P., Tony Augustine, Machine Learning Approach on Degree-Sum Based Topological Indices of Bottom-up-Synthesized Graphene Nanoribbons, *Nano*, 2025, 2650008. **DOI**: 10.1142/S1793292026500086.
7. Abdelli Fathi, Tony Augustine, Mohamad Nazri Husin, **Sahaya Vijay Jeyaraj**, Jain Maria Thomas, Roy Santiago, Graph Theoretical Approach to Topological Features of Zeolitic Tetrahedral Imidazolate Framework-3, *Malaysian Journal of Fundamental and Applied Sciences (MJFAS)*, 21(2), 2025. **DOI**: 10.11113/mjfas.v21n2.4127.
8. **Sahaya Vijay Jeyaraj**, Roy Santiago, Chandra Sekar Ponnusamy, Jain Maria Thomas, Mohamad Husin, Abdelli Fathi, Efficient Generation of Specific Counting Polynomials and Its Subsequent Indices of Complex Boric Acid Structure, *Current Organic Synthesis*, 22(7) 2025, 821-828. **DOI**: 10.2174/0115701794366590250414051549.
9. Tony Augustine, Roy Santiago, **Sahaya Vijay Jeyaraj**, Mohamad Azeem, Exploring Neighborhood Topological Descriptors for Quantitative Structure-Property Relationship (QSPR) Analysis and Entropy Measures of Some Anti-cancer Drugs, *Current Organic Synthesis*, 22(8), 2025, 934-943. **DOI**: 10.2174/0115701794349166241217085334.
10. **Sahaya Vijay Jeyaraj**, R. Santiago, M. Azeem, Tony Augustine, M. N. Husin, Quantitative Structure-Property Relationship (QSPR) Modeling of Central Nervous System (CNS) Drug Activity Using Molecular Descriptors, *Current Organic Synthesis*, 22(7), 2024, 799-810. **DOI**: 10.2174/0115701794349717241024100059.
11. A. Fathi, **Sahaya Vijay Jeyaraj**, M. N. Husin, Tony Augustine, Valency-Based Molecular Descriptor on Structural Property Relationship of Ni Tetrathiafulvalene Tetrathionate, *Malaysian Journal of Fundamental and Applied Sciences (MJFAS)*, 20(6), 2024. **DOI**: 10.11113/mjfas.v20n6.3831.
12. Sonadevi S., Rajaraman D., Periyasamy D., **Sahaya Vijay Jeyaraj**, Tony Augustine, Roy Santiago, Saritha M., Athishu Anthony L., In Silico Rationalization of Piperidin-4-One Analogs as Anaplastic Lymphoma Kinase Inhibitors: Integration of Density Functional Theory, Topological Descriptor, Molecular Docking and Pharmacological Feature Profiling, *Journal of Molecular Structure*, 1311, 2024, 138356. **DOI**: 10.1016/j.molstruc.2024.138356.

13. **Sahaya Vijay Jeyaraj**, Roy Santiago, A Study on Efficient Technique for Generating Vertex-based Topological Characterization of Boric Acid 2D Structure, *ACS Omega*, 8(25), 2023, 23089–23097. **DOI**: 10.1021/acsomega.3c02477.
14. **Sahaya Vijay Jeyaraj**, Roy Santiago, Charles Beromeo Bheeter, Mohamad Nazri Husin, Tony Augustine, Rudrusamy U. Gobithaasan, Easuraja Michael, Topological Properties and Entropy Calculations of Aluminophosphates, *Mathematics*, 11(11), 2023, 2443. **DOI**: 10.3390/math11112443.
15. **Sahaya Vijay Jeyaraj**, Roy Santiago, Computation of Wiener Descriptor for Melamine Cyanuric Acid Structure, *Polycyclic Aromatic Compounds*, 44(2), 2023, 1057–1071. **DOI**: 10.1080/10406638.2023.2186441
16. Tony Augustine, Roy Santiago, **Sahaya Vijay Jeyaraj**, Jain Maria Thomas, Shanmugam Palanisamy, Topological study on degree based molecular descriptors of fullerene cages, *Molecular Physics*, 121(4), 2023, e2179858. **DOI**: 10.1080/00268976.2023.2179858.

Invited Talk

1. **Sahaya Vijay Jeyarj,** Resource Person at **"**Mathematical Tools and Modern Techniques for Research, Thesis Writing and Scholarly Publication" at Dhanalakshmi Srinivasan University, Tiruchirappalli, Tamil Nadu, India on **September 8th, 2026.**
2. **Sahaya Vijay Jeyarj,** International invited speaker at **"**International Faculty Development Programme (FDP) on Enhancing Research Capacity: Publication, Grants and Research Impact." at Dhanalakshmi Srinivasan University, Tiruchirappalli, Tamil Nadu, India on **August 27th, 2026.**
3. **Sahaya Vijay Jeyarj**, Keynote speaker at "3rd International Conference on Recent Trends in Data Science (ICRTDS-2026)" at St. Xavier's College (Autonomous), Palayamkottai, Tamil Nadu, India on **March 5-March 6, 2026.**
4. **Sahaya Vijay Jeyarj,** Keynote International speaker at **"**3rd International Conference on Innovation in Science and Humanities (ICISH-2025)" at Saveetha Engineering College in Chennai, Tamil Nadu, India on **August 19-Aug 20, 2025.**

CONFERENCE (SELECTED)

1. **Sahaya Vijay Jeyarj,** Hirokazu Komatsu, Hideshi Ooka, Ryuhei Nakamura, "Asian Conference for Mathematical Biology", Kyoto Terrsa, Kyoto, Japan, **July 11- Jul 17, 2025.**
2. **Sahaya Vijay Jeyaraj**, "Quantitative-Structure Activity Relationship Based Drug Designing" BIRAC - BioNEST Laboratory, VIT-TBI, VIT University, Vellore, India, **March 11, 2023.**
3. **Sahaya Vijay Jeyaraj**, "One Week International FDP on Advanced Tools and Techniques for Scientific Research Writing and Publication" Department of Mathematics and Statistics, University of Engineering and Management (UEM), Jaipur, India, **September 12-18, 2022**.
4. **Sahaya Vijay Jeyaraj**, "International Conference on Recent Developments in Mathematics" Faculty of Engineering, Applied Science and Technology, Canadian University Dubai, UAE, **August 24-26, 2022**.

5. **Sahaya Vijay Jeyaraj**, "International Webinar on Recent Trends in Pure and Applied Mathematics" Dept of Mathematics, The Gandhigram Rural Institute (Deemed to be University), Dindigul, Tamil Nadu, India, **June 26-27, 2022**.
6. **Sahaya Vijay Jeyaraj**, "A Webinar on Mathematics in Data Science" Dept of Mathematics (Shift-I), Patrician College of Arts and Science, Chennai, Tamil Nadu, India, **June 25, 2021**.

ORAL PRESENTATIONS (SELECTED)

1. **Sahaya Vijay Jeyaraj,** Hirokazu Komatsu, Hideshi Ooka, Ryuhei Nakamura, *Calculating the lifetime of autocatalytic networks* at "Joint conference of Asian Conference of Mathematical Biology and Annual Meetings of Japanese Society for Mathematical Biology", Kyoto Terrsa, Kyoto, Japan, **July 11- Jul 17, 2025.**
2. Hideshi Ooka, Hirokazu Komatsu, **Sahaya Vijay Jeyaraj**, Ryuhei Nakamura *Timescale Decomposition of Linear Chemical Reaction Networks* "Asian Conference for Mathematical Biology", Kyoto Terrsa, Kyoto, Japan, **July 11- Jul 17, 2025.**
3. Hideshi Ooka, **Sahaya Vijay Jeyaraj**, Ryuhei Nakamura *Analyzing the Stability of Autocatalytic Chemical Reaction Networks Based on their Amplification Rate*, "Japan Geoscience Union Meeting 2023", Makuhari Messe, Chiba, Japan, **May 29, 2025.**
4. **Sahaya Vijay Jeyaraj**, A *Cut Method Technique Used on Topological Descriptors of Melamine Cyanuric Acid*, "International Conference on Mathematics and Its Applications (ICMA -2024)" at Sacred Heart College (Autonomous), Tirupattur, India, **February 15-16 (2024)**.
5. **Sahaya Vijay Jeyaraj** and Roy Santiago, *Topological Properties and Entropy Calculations of Aluminophosphates*, "International Conference on Modelling, Simulation and Optimization of Energy Systems (MSOES-2023)", Canadian University Dubai, UAE, **June 17-18 (2023)**.
6. **Sahaya Vijay Jeyaraj** and Roy Santiago, *Computation of Wiener Descriptor for Melamine Cyanuric Acid Structure*, "International Conference on Recent Developments in Mathematics (ICRDM -2022)", **August 24-26 (2022)**, Canadian University Dubai, UAE.

POSTER PRESENTATION

1. **Sahaya Vijay Jeyaraj,** Ryuhei Nakamura, Hideshi Ooka, *Calculating the Lifetime of Autocatalytic Chemical Reactions Network*, "CSRS Interim Progress Report Meeting, RIKEN, Yokohama Campus", Yokohama, Japan, **November 17 (2025)**.

WORKSHOP

1. iTHEMS (RIKEN)-Theory Center (KEK) Joint Workshop **2025**: Scientific Writing and Diversity-Equity-Inclusion, RIKEN, Japan.

RESEARCHER ID

1. **Google Scholar ID**: https://scholar.google.com/citations?hl=en&user=iJlA0VQAAAAJ
2. **ORCID**: 0000-0001-6672-0092
3. **Sci-Profiles**: 2310023
4. **Scopus Author ID**: 58404666100

5. **Web of Science Researcher ID**: AIA-9120-2022

REFERENCES

1. Dr. S. Roy,
   Assistant Professor (Senior Grade II),
   Department of Mathematics,
   Vellore Institute of Technology (VIT University),
   Vellore – 632014, Tamil Nadu, India.
   Email: roy.santiago@vit.ac.in
2. Prof. Ryuhei Nakamura,
   Team Leader,
   CSRS Biofunctional Catalyst Research Team,
   RIKEN,
   Wakoshi, Saitama 351-0106, Japan.
   Email: ryuhei.nakamura@riken.jp
3. Dr. Hideshi Ooka,
   Team Leader,
   Mathematical Catalysis Research Team,
   Research Center for Energy and Environmental Materials (GREEN),
   National Institute for Materials Science
   Email: ooka.hideshi@nims.go.jp